

Curriculum Vitae: Dr. Violeta Damjanovic-Behrendt



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Professional Experience

Current Position:

GreenTwin GmbH

- January 2022 – present: CTO & Principal Scientist at GreenTwin GmbH

Contracts with the European Commission:

Independent Expert/ Horizon Europe Programme (including MSCA)

- January 2021 - present

Independent Expert/ H2020 Research and Innovation Programme

- January 2016 – December 2020

Past Positions:

Salzburg Research
Forschungsgesellschaft, Salzburg,
Austria

- June 2021 - February 2024: Senior Researcher, Intelligent Connectivity group (*part time*)
- February 2014 – June 2021: Senior Researcher, Internet of Things (IoT) group (*full time*)
- November, 2006 – January 2014: Senior Researcher at the Knowledge and Media Technologies (KMT) group (*full time*)

Postal Savings Bank, Belgrade, Serbia

- February, 2004 – December, 2006: *Software Project Manager (e-banking services) (full time)*
- April, 2002 – February, 2004: *Programmer Analyst (PL/I, DB2, CICS, Visual Basic, OS/390, Java) (full time)*
- December, 1998 – April, 2002: *Application programmer (PL/I, DB2, CICS, Visual Basic, operating system OS/390) (full time)*

Past Teaching Experience (External lecturer):

Salzburg University of Applied
Science, Austria

- July 2013 – (BSc course) **Game Theory for the Smart Grid** (*part time*)

Salzburg University of Applied
Science, Austria

- October / November 2011 – (Master course, MultiMedia Technology (MMT) Department) **Artificial Intelligence and Algorithmic Game Theory** (*part time*)

Degrees & Thesis

- (January 2006 – November 18, 2008): *PhD in Information Systems*, Department of Information Systems and Technologies, FON, University of Belgrade, Serbia, supervised by Prof. Dr. Vladan Devedzic.

“Ambient Intelligence in Adaptive Online Experiments”, PhD thesis, University of Belgrade, Serbia (in Serbian), 2008 (fields: **Ambient Intelligence; Personalized Adaptation; Multi Agent Systems (MAS); Technology Enhanced Learning (TEL)**)

- (October 2003 – September 21, 2005): *MSc in Electronic Business*, Department of Information Systems and Technologies, FON, University of Belgrade, Serbia, supervised by Prof. Dr. Vladan Devedzic.

“Web Pedagogical Agents: A Touch of Web Intelligence”, MSc thesis, University of Belgrade, September Serbia, 2005 (fields: **Multi Agent Systems (MAS); Technology Enhanced Learning**)

- (October 2001 – October 07, 2003): *Degree as a Specialist in Internet Technologies*, Department of Information Systems and Technologies, FON, University of Belgrade, Serbia, supervised by Prof. Dr. Vladan Devedzic.

“Semantic Web, Ontologies and Agents”, Work for the Degree as a Specialist in Internet Technologies, University of Belgrade, Serbia, 2003 (fields: **Semantic Web; Web Multi Agent Systems (MAS)**)

- (September 1994 – October 27, 1998): *BSc (Dipl. Ing.) in Information Systems*, Department of Information Systems and Technologies, FON, University of Belgrade, Serbia, supervised by Prof. Dr. Milan Martic and Prof. Dr. Slobodan Krcevinac.

“Using Data Envelopment Analysis (DEA) to Evaluate Investments Projects”, BSc (Dipl. Ing.) thesis, University of Belgrade, 1998 (field: **Operations Research; Linear Programming**)

Computer Skills

- **Digital systems & architecture:** Modular system design for distributed enterprise systems; Service-oriented architectures (SOA) and workflow orchestration (BPEL, WSDL); UML-based system modelling and governance-driven architecture design; Knowledge graph modelling and semantic interoperability frameworks
- **Semantic & compliance technologies:** Ontology engineering (Owl, RDF), SPARQL querying and semantic data integration; XML-based data exchange models; Governance-aware data modelling for interoperability

- **AI & intelligent systems:** Knowledge graph-enhanced reasoning systems; LLM/SLM conceptual integration in enterprise architectures; Semantic harmonisation and rule-based validation logic
- **Modern AI & SLM operationalization:** HuggingFace ecosystem; Transformers; Fine-tuning small models (LoRA); Vector databases (FAISS, Weaviate, Pinecone); RAG architectures
- **Data engineering & databases:** SQL (DB2, MySQL, PostgreSQL); ORM-based data abstraction; Structured data integration in enterprise environments
- **Data governance & interoperability standards:** JSON-LD; SHACL validation; DCAT-AP; Data spaces (IDS, Gaia-X basics); API security (OAuth2, OIDC)
- **Cloud-native integration environments:** Docker & Kubernetes; CI/CD pipelines (GitHub Actions); API-first design (OpenAPI, RESTful microservices); data pipelines (Airflow)
- **Geospatial & Earth Observation processing:** Python geospatial stack (GeoPandas, GDAL), PostGIS, STAC standard; Copernicus Sentinel data handling; Basic QGIS and Google Engine
- **Programming & analytics:** Python (data processing, backend services); Java (enterprise web systems); JavaScript (Next.js); R (statistical modelling); C/C++, Visual Basic, Pascal, Cobol

Sustainability Skills

- GHG Protocol accounting framework
- Python libraries for carbon accounting; ESG reporting data structures (CSRD alignment); KPI dashboard frameworks (Plotly, Power BI, Superset)

Research Interests

- Artificial Intelligence (explainable AI, GenAI, casual reasoning, LLM, SLM), Game Theory
- Machine Learning and Data Science for Earth Observation, GeoAI
- Semantic Web, Ontology Engineering, and reasoning
- Digital Twin modelling

Research Awards

- August 2020: Best Paper Award for the research paper: *W. Behrendt and V. Damjanovic-Behrendt, "Establishing and Maintaining Situation Awareness for Shop-Floor Assistance in Manufacturing", In Proceedings of the 10th IEEE Conference on Cognitive and Computational Aspects of Situation Management (CogSIMA 2020), University of Victoria, British Columbia, Canada.*
- April 2018: *The Most Innovative Solution Award for the research paper: "An Ontology Framework for Multisided Platform Interoperability"* (I-ESA 2018 conference, Berlin, Germany)
- November 10, 2005: *The 2005 Research Award - Serbian Society for Informatics* (MSc thesis: "Web Pedagogical Agents: A Touch of Web Intelligence"), Serbian Chamber of Commerce, Belgrade, Serbia

Research Projects

Projects with GreenTwin GmbH (January 2024 - present)

HE ResC4EU
January 2024 –
December 2026

ResC4EU (Resilient Supply Chains for Europe)
HORIZON-CL4-2023-RESILIENCE-01

- **Project website:** <https://www.resc4eu.com/>

FFG CE-PASS

January 2022 –
December 2023

CE-PASS (Circular Economy - Digital Product Passport)
FO999887526

H2020-DT-EFPF

January 2019 –
December 2022

EFPF (European Connected Factory Platform for Agile Manufacturing)

- **Project website:** <http://efactory-project.eu/>

H2020-FoF-NIMBLE

October 2016 –
September 2019

NIMBLE (Collaborative Network for Industry, Manufacturing, Business and Logistics in Europe)

- **Project website:** <https://www.nimble-project.org/>

FFG/bmvit

IoT4CPS

December 2017 –
November 2020

IoT4CPS (Trustworthy IoT for CPS)

- **Project website:** <https://iot4cps.at/>

IoT4Industry

September 2014 –
August 2015

- **Project website:** <https://iot4industry.salzburgresearch.at/>

EAGLE

February 2014 –
January 2017

EAGLE (EnhAnced Government LEarning)

- **Project website:** <http://www.accelopment.com/en/projects/eagle>

**FFG Sondierung
UNDERSTANDER**

September 2013 –
August 2014

- **Project website:**
<http://www.salzburgresearch.at/en/projekt/understander-2/>

OP4L

January 2011 –
September 2013

OP4L (Online Presence for Learning)

IKS

January 2009 –
December 2012

IKS (Interactive Knowledge Stack)

ImportNET

November 2006 –
September 2009

ImportNET (Intelligent modular open source platform for intercultural and cross- domain SME networks)

- **Project website:** <http://www.importnet-project.org/>

GRISINO

November 2006 –
September 2008

GRISINO (GRID Semantics and Intelligent Objects).

- **Project website:** <http://grisino.salzburgresearch.at/>
<http://www.grisino.at/>

ProLearn

October 2004 –
October 2006

- **ProLearn** “Network of Excellence” integrates the key areas of research most relevant to professional learning in SMEs and larger companies. It aims at bridging the gap between research and education between research and education at university and learning at the workplace.

The GOOD OLD AI research group was an associated partner in the ProLearn project. My research was on the topic of personalized adaptive learning (WP1).

Project home: <http://www.prolearn-project.org/>

Recent publications (selected short list of publications) _____

Book Chapters:

1. R.C. Sofia, C. Coutinho, G. Scivoletto, G. Insolubile, R.A. Deshmukh, A. Schneider, V. **Damjanovic-Behrendt**, F. Gigante, U. Wajid, A. Nizamis, D. Ioannidis, T. Mastos. “The EFPP approach to manufacturing applications across Edge-Cloud”. In R. C. Sofia, J. Soldatos (Editors), *Shaping the Future of IoT with Edge Intelligence – How Edge Computing Enables the Next Generation of IoT Applications*. River Publishers, 2023.

International Journals:

2. R.A. Deshmukh, D. Jayakody, A. Schneider, V. **Damjanovic-Behrendt**, Data Spine: A Federated Interoperability Enabler for Heterogeneous IoT Platform Ecosystems. *Sensors* 2021, 21, 4010. Special Issue “Industry 4.0 and Smart Manufacturing” (impact factor 2019: 3.275). <https://doi.org/10.3390/s21124010>
3. V. **Damjanovic-Behrendt**, W. Behrendt, “An Open Source Approach to the Design and Implementation of Digital Twins for Smart Manufacturing”. *International Journal of Computer Integrated Manufacturing*. Special Issue on Cyber Physical Systems with Applications in Production and Logistics. 2019. [Taylor & Francis](https://doi.org/10.1080/0951192X.2019.1599436) (impact factor 2018: 2.861). Published on Taylor & Francis
Online: <https://www.tandfonline.com/doi/abs/10.1080/0951192X.2019.1599436>
4. T. Kurz, G. Güntner, V. **Damjanovic**, S. Schaffert, M. Fernandez, “Semantic Enhancement for Media Asset Management Systems: Integrating the Red Bull Content Pool into the Web of Data”, *Multimedia Tools and Applications (MTAP)* (2012), doi: 10.1007/s11042-012-1197-7, (Impact factor 2010: 0.885)
5. V. **Damjanovic**, “Semantic Reengineering of Business Processes “, *Information Systems* (2009), *Special Issue on Vocabularies, Ontologies, and Rules for Enterprise and Business Process Modelling and Management*, doi: 10.1016/j.is.2009.06.003, (ISSN: 0306-4379; Impact factor 2008: 1.660)

International Conferences & Workshops:

2026-2018(an excerpt)

6. V. **Damjanovic-Behrendt**, A. Kinra, "eFTI-enabled Sustainable Intelligence for Supply Chain and Logistics Resilience and Decarbonization". **Submitted** to EurOMA 2026, Vienna, Austria, June 27th-July 2nd 2026: <https://www.euroma2026.org/>

7. **V. Damjanovic-Behrendt**, D. Bhardwaj, A. Kinra, "eFTI-enabled Multi-Sided Supply Chain Platforms: Design Principles and Practical Extensions for Bridging eFTI Regulatory Compliance, Supply Chain Resilience, Sustainability, and Decarbonization Targets". **To be presented** at the *World Conference of Transport Research* (WCTR 2026), Toulouse, 2026
8. **V. Damjanovic-Behrendt**, "Leveraging Earth Observation Data for Robust Supply Chain Resilience". Poster presented at the ESA's Living Planet Symposium 2025, Session F.05.01 – Satellite EO data benefit the economy, the environment and our societies: the evidence and the stories, June 22-27, 2025, Vienna, Austria. Available from: <https://zenodo.org/records/15750587>
9. **V. Damjanovic-Behrendt** and W. Behrendt, "Digital Solutions for Resilient and Regenerative Supply Chains", In Proceedings of the 12th EurOMA (Sustainable Operations and Supply Chains) Forum 2025, Track on: Regenerative operations and supply chains for sustainable positive impact, March 24-27, 2025. Available from: <https://zenodo.org/records/15750538>
10. O. Veledar, E. Armengaud, L. Happ Botler, **V. Damjanovic-Behrendt**, C. Derler, S. Jaksic, L. Krammer, C. Lettner, G. Macher, S. Marksteiner, A. Martin, M. Matschnig, P. Priller, S. Ramacher, K. Roemer, C. Schmittner, C. Tiefnig, H. Vallant, H. Weiskirchner, and M. Drobics, "Steering Drivers of Change: Maximizing Benefits of Trustworthy IoT". In Proceedings of the 28th European System, Software and Service Process Improvement & Innovation Conference (EuroSPI2021), September 01-03, 2021, Krems, Austria.
11. C. Schranz, F. Strohmeier, and **V. Damjanovic-Behrendt**, "A Digital Twin Prototype for Product Lifecycle Data Management", In Proceedings of the IEEE International Symposium on Cyber Physical Systems Concerns and Applications (IEEE CPS'CAPP 2020), in conjunction with the 17th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA 2020), November 02-05 2020, Antalya, Turkey.
12. W. Behrendt and **V. Damjanovic-Behrendt**, "Establishing and Maintaining Situation Awareness for Shop-Floor Assistance in Manufacturing", In Proceedings of the 10th IEEE Conference on Cognitive and Computational Aspects of Situation Management (CogSIMA 2020), August 2020, University of Victoria, British Columbia, Canada (**BEST PAPER AWARD**).
13. **V. Damjanovic-Behrendt**, W. Behrendt, "An Open Source Approach to the Design and Implementation of Digital Twins for Smart Manufacturing". International Journal of Computer Integrated Manufacturing. Special Issue on Cyber Physical Systems with Applications in Production and Logistics. 2019. Taylor & Francis, doi: <https://doi.org/10.1080/0951192X.2019.1599436> (impact factor 2017: 1.995).
14. D. Jayakody, N. Selvanathan and **V. Damjanovic-Behrendt**, "Federated Search and Recommendation", In Proceedings of the Workshop on Digital Platform Ecosystems: From Interoperability to Federation (IFED 2020), in conjunction with the I-ESA 2020 conference, March 24-25, 2020, Tarbes, France.
15. N. Selvanathan, D. Jayakody and **V. Damjanovic-Behrendt**, "Federated Identity Management and Interoperability for Heterogeneous Cloud Platform Ecosystems", In Proceedings of the 14th International Conference on Availability, Reliability and Security

- (ARES'19), Workshop on Industrial Security and IoT (WISI 2019), August 26-29, 2019, Canterbury, UK. (CORE2018 B) Online: <https://doi.org/10.1145/3339252.3341492>
16. O. Veledar, **V. Damjanovic-Behrendt**, G. Macher, "Digital Twins for Dependability Improvement of Autonomous Driving". In Proceedings of the Workshop on Best Practices in Design of Systems Applying Functional Safety & Cybersecurity (co-located with the 26th European System, Software and Service Process Improvement & Innovation Conference (EuroSPI2019)), September 18-20, 2019, Edinburgh, Scotland, UK. (CORE2018 B) Online: <http://2019.eurospi.net/index.php/workshop?id=77>
 17. **V. Damjanovic-Behrendt**, "A Digital Twin-based Privacy Enhancement Mechanism for the Automotive Industry", in Proceedings of the 9th International Conference on Intelligent Systems: Theory, Research and Innovation in Applications. September 25-27, 2018, Madeira, Portugal.
 18. J. Innerbichler, **V. Damjanovic-Behrendt**, "Federated Byzantine Agreement to Ensure Trustworthiness of Digital Manufacturing Platforms", In Proceedings of the 1st Workshop on Cryptocurrencies and Blockchains for Distributed Systems (CryBlock 2018) (co-located with the 16th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys 2018)) Munich, Germany. Online: <https://dl.acm.org/citation.cfm?id=3211953>
 19. **V. Damjanovic-Behrendt**, "Cybersecurity Requirements for Supporting Enterprise Interoperability of Multi-Sided Platforms", In Proceedings of the 9th International Conference on Interoperability for Enterprise Systems and Applications (I-ESA 2018), Berlin, Germany, March 22-23, 2018.
 20. Q. Deng, S. Gönül, Y. Kabak, N. Gessa, D. Glachs, F. Gigante, **V. Damjanovic-Behrendt**, K. Hribernik, K.D. Thoben, "An Ontology Framework for Multisided Platform Interoperability", In Proceedings of the 9th International Conference on Interoperability for Enterprise Systems and Applications (I-ESA 2018), Berlin, Germany, March 22-23 2018.
The Most Innovative Solution Award.

Invited speaker

- AVL Research Networking Day 2019, Graz, 19 November 2019. Presentation of an invited poster on the topic "Digital Twin-based Traceability of Product Data Lifecycle". For more details: <https://violetabehrendt.com/2019/11/23/at-the-avl-research-networking-day-2019-graz/>
- Vienna CyberSecurity Week 2019, 14 March 2019, at the Austrian Federal Economic Chamber (WKO), Vienna, Austria. Talk: "Digital Twins for Automotive Cybersecurity Engineering and Validation" at the session "Cyber Security in Industry". Online: https://www.energypact.org/wp-content/uploads/2019/03/VCSW-Conference-Programme_12.pdf
- Community session 3-in-1 at the ICT 2018: Imagine Digital, 5 December 2018, Austrian Center Vienna. Talk "Digital Twin for Safety and Security in the Automotive Industry" as part of the IoT4CPS community session. Online: <https://iot4cps.at/2018/12/10/iot4cps-ict-2018-imagine-digital/>
- February 24, 2009 "–Ontologizing Business Processes" – Research seminar of *Process Modelling Patterns*, Institute of Cognitive Sciences and Technologies (ISTC-CNR), Semantic Tech. Lab. (STLab), Rome, Italy
- May 11, 2004 "–The Semantic Web", *Symposium on Informatics 2004*, Belgrade fair, Belgrade, Serbia